

Capacity-Limited Cosmology from Informational Bounds in Quantum Hodon Incidence Structures

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Abstract

The Boltzmann Brain (BB) problem challenges the coherence of cosmological models in which extremely rare entropy-lowering fluctuations eventually dominate over ordinary observers. Recent finite-dimensional proposals suppress BB formation dynamically by restricting the universe to a periodic, non-ergodic orbit in Hilbert space. In this work we develop a more structural resolution within the Quantum Hodon Incidence Structure (QHIS), a discrete, operationally defined substrate of elementary informational events referred to as hodons. QHIS models impose a holographic-style capacity bound on every operational region, ensuring that its accessible informational content is finite and determined solely by the combinatorial properties of its incidence boundary. When combined with a commensurable-spectrum global Hamiltonian, this constraint yields a cosmology that is periodic, non-ergodic, and fundamentally capacity-limited: ordinary observers fit within the modest informational budgets of suitable regions, whereas Boltzmann-Brain-level observers require correlation structures far exceeding the capacity of any region and are therefore excluded in principle.

We construct explicit toy models illustrating (i) minimal observers supported by small operational regions, (ii) Page-curve-like behavior arising from finite correlation capacity across incidence boundaries, and (iii) a small- M periodic QHIS cosmology in which no region attains the complexity required for BB formation. These results suggest that informational granularity, rather than geometric volume or temporal duration, determines the space of physically realizable observers. From a quantum-gravity perspective, the QHIS capacity bound provides a discrete, pre-geometric analogue of holographic entropy constraints and may serve as a structural precursor to generalized entropy in the QES/island framework. In such universes, ordinary observers arise naturally, while Boltzmann Brains are excluded by the fundamental informational structure of the world.

Keywords: Quantum gravity; holographic entropy bounds; informational capacity; Boltzmann Brains; Page curve; discrete pre-geometric models; commensurable-spectrum Hamiltonian; non-ergodic evolution; incidence structures; operational regions; correlation capacity; emergent geometry; generalized entropy; quantum extremal surfaces.

1 Introduction

The Boltzmann Brain (BB) problem is a longstanding conceptual challenge in cosmology [1, 2, 3, 4]. In models containing a long-lived or eternal de Sitter phase [5], rare entropy-lowering fluctuations

can, in principle, produce self-aware structures with extremely small probability but over extremely long timescales. If the spacetime volume or duration of the de Sitter epoch is sufficiently large, such fluctuations dominate over ordinary observers, leading to the paradoxical conclusion that the “typical observer” is a spontaneously assembled, disordered brain rather than an evolved, environmentally embedded one [6]. This undermines the empirical coherence of cosmology, since a BB-dominated universe would render our observational situation atypical [7].

A recent proposal by Carroll, Diachenko, and Dulani [8] identifies a dynamical loophole within finite-dimensional quantum cosmology. If the global Hilbert space is finite and the Hamiltonian spectrum is *commensurable* [9], the unitary evolution becomes exactly periodic: the universe traces a one-dimensional, non-ergodic orbit in Hilbert space, revisiting the same sequence of macrostates every period. In such a model, the effective duration of any high-entropy epoch is bounded by the period $T_\star$, and BB formation is suppressed provided $T_\star \ll \Gamma_{\text{BB}}^{-1}$, where Γ_{BB} is the BB nucleation rate. This mechanism limits the dynamical exploration of Hilbert space without modifying the underlying physics of fluctuations.

In this work we show that, within the Quantum Hodon Incidence Structure (QHIS) framework [10], a sharper and more structural resolution of the BB problem becomes available. QHIS models describe the universe as a discrete, operationally grounded network of elementary informational events (“hodons”) connected by directed links with probabilistic causal weights. Every operational region R possesses an incidence boundary ∂R , and the informational content accessible to observers confined to R is limited by a holographic-style counting principle: the capacity of a region is determined not by its volume or temporal duration, but by the combinatorial structure of its boundary. This constraint is independent of geometry and applies even in non-geometric or pre-geometric regimes. In particular, it implies that no region of the universe can support more than a finite amount of structured information, regardless of how long it persists.

A central theme of this work is that Page-curve-like behavior [11, 12] arises naturally in QHIS from finite correlation capacity across incidence boundaries. In semiclassical gravity, the Page curve reflects the inability of a region to sustain arbitrarily large entanglement: correlations grow, saturate at a boundary-determined maximum, and then purify [13]. We show that the same qualitative behavior emerges in QHIS without reference to geometry or quantum extremal surfaces. The Page-curve-like profile thus becomes an operational signature of finite informational capacity, and the mechanism that limits entanglement growth also limits the complexity of observers. In this way, the capacity-limited Page curve and the structural exclusion of Boltzmann Brains arise from a single underlying principle.

When a commensurable-spectrum Hamiltonian is imposed on the global QHIS Hilbert space, the resulting cosmology is not only periodic and non-ergodic, but also *capacity-limited*. Ordinary observers, whose informational complexity is modest, can be supported within the finite capacities of suitable regions. In contrast, Boltzmann Brain-level observers require correlation structures far exceeding the holographic capacity of any operational region [14]. In a capacity-limited QHIS cosmology, BBs are therefore not merely improbable—they are *structurally impossible*. The BB problem is reframed as a question about the granularity of observers permitted by the informational substrate of the universe.

To make this argument concrete, we construct explicit toy models. We define a minimal observer as a structured pattern over a small number of hodons with a few bits of sensory and memory capacity, and show that such observers fit within the holographic capacity of small regions. Conversely, BB-level observers require informational capacities orders of magnitude larger than any region can support. We then build a periodic QHIS cosmology with a commensurable-spectrum Hamiltonian and a capacity-limited Page curve, illustrating how black-hole evaporation, entropy saturation, and purification arise without reference to semiclassical geometry or quantum extremal

surfaces (QES) [15].

Beyond its cosmological implications, the QHIS framework is naturally situated within discrete and information-theoretic approaches to quantum gravity. Its incidence structure parallels the causal-order foundations of causal set theory [16, 17], while its boundary-based capacity constraint echoes the area-law scaling of holographic entropy bounds [18, 19]. The finite correlation capacity across incidence boundaries is reminiscent of tensor-network models of emergent geometry, where entanglement structure determines effective spatial relations [20]. These connections suggest that QHIS may serve as a pre-geometric substrate from which semiclassical spacetime and holographic behavior emerge under coarse-graining.

2 Background

This section reviews the three ingredients on which the present work builds: (i) the Boltzmann Brain problem in semiclassical cosmology, (ii) Carroll’s commensurable-spectrum proposal for a finite-dimensional, exactly periodic universe, and (iii) the Quantum Hodon Incidence Structure (QHIS) framework, which provides a discrete, operational foundation for information flow and emergent geometry. These elements supply the conceptual tools needed for the capacity-limited, periodic cosmology developed in later sections.

2.1 Boltzmann Brains in semiclassical cosmology

In standard Λ CDM cosmology, the late-time universe approaches a quasi-de Sitter phase with horizon temperature T_{dS} and finite entropy S_{dS} . Quantum and thermal fluctuations in this background can, in principle, generate highly ordered structures with extremely small probability. Among these, the most conceptually troubling are *Boltzmann Brains* (BBs): spontaneously assembled, self-aware subsystems arising from rare entropy-lowering fluctuations. Although the probability of forming a BB is exponentially suppressed, the de Sitter epoch persists for exponentially long timescales. If the product of duration and volume is sufficiently large, the expected number of BBs exceeds the number of ordinary observers produced by cosmological evolution.

This leads to the well-known paradox: in a sufficiently long-lived de Sitter universe, the “typical observer” is a BB rather than an evolved, environmentally embedded observer. Such a universe is empirically incoherent, since BBs possess disordered memories and atypical observational histories. Avoiding BB domination has therefore become a consistency requirement for any viable cosmological model.

2.2 Carroll’s commensurable-spectrum loophole

Carroll et al. recently introduced a method in [8] to prevent BB domination in a finite-dimensional quantum cosmology. The key idea is to impose a *commensurable* energy spectrum on the global Hamiltonian H , such that all energy differences are integer multiples of a fundamental frequency. Under this condition, the unitary evolution becomes exactly periodic:

$$U(T_\star) = e^{-iHT_\star} = \mathbb{I}, \quad (2.1)$$

for some finite period $T_\star$. The universe evolves along a one-dimensional, non-ergodic orbit in Hilbert space, revisiting the same sequence of macrostates every cycle.

In this setting, the effective duration of any high-entropy epoch is bounded by $T_\star$. If $T_\star$ is chosen to satisfy $T_\star \ll \Gamma_{\text{BB}}^{-1}$, where Γ_{BB} is the BB nucleation rate, then the expected number of BBs per cycle is negligible. BBs are not forbidden, but they are dynamically suppressed by the periodic,

non-ergodic structure of the evolution. This provides a loophole to the usual BB argument, but it remains a *dynamical* solution: BBs are allowed in principle, yet the universe never evolves long enough in any direction of Hilbert space for them to dominate.

2.3 The QHIS framework

The Quantum Hodon Incidence Structure (QHIS) framework provides a discrete, operationally grounded model of information flow on a pre-geometric substrate. Its mathematical structure is closely aligned with several approaches to quantum gravity in which causal relations and finite information content play foundational roles. The directed incidence relations of a QHIS parallel the causal order of causal set theory, while the assignment of finite-dimensional Hilbert spaces to links resembles spin-network and tensor-network constructions. The holographic density B plays a role analogous to the Bousso bound or covariant entropy bound, but formulated in purely combinatorial terms without reference to geometric area. This makes QHIS compatible with both pre-geometric and semiclassical regimes, and positions it naturally within the broader landscape of discrete quantum gravity models.

2.3.1 Formal structure of a QHIS

A QHIS models the informational degrees of freedom associated with a discrete causal substrate. The basic ingredients are a set of elementary informational events called *hodons*, a set of directed incidence relations, and finite-dimensional Hilbert spaces assigned to each. The following definitions make this precise.

Definition 1 (QHIS). *A QHIS is a triple*

$$\mathcal{Q} = (P, L, \mathcal{H}), \quad (2.2)$$

where:

1. P is a finite or countable set of *hodons* (elementary informational events).
2. $L \subseteq P \times P$ is a set of *directed incidence relations* (links), equipped with non-negative causal weights $\omega : L \rightarrow \mathbb{R}_{\geq 0}$.
3. $\mathcal{H}$ assigns to each hodon $p \in P$ a finite-dimensional Hilbert space $\mathcal{H}_p$, and to each link $\ell \in L$ a finite-dimensional Hilbert space $\mathcal{H}_\ell$ that represent intrinsic informational degrees of freedom of p and ℓ , correspondingly.

The global Hilbert space of the QHIS is

$$\mathcal{H}_{\text{global}} = \bigotimes_{p \in P} \mathcal{H}_p \otimes \bigotimes_{\ell \in L} \mathcal{H}_\ell. \quad (2.3)$$

Operational regions play a central role in defining observers and capacity bounds. Their boundaries are determined purely by incidence relations.

Definition 2 (Operational Region). *A subset $R \subseteq P$ is called an operational region. Its incidence boundary ∂R is the minimal set of links*

$$\partial R = \{\ell = (p, q) \in L : p \in R, q \notin R \text{ or } p \notin R, q \in R\}, \quad (2.4)$$

i.e. the set of directed links whose removal disconnects R from its complement.

To quantify the informational capacity of a region, we introduce a boundary measure and a holographic density.

Definition 3 (Boundary Measure). *Let $v : L \rightarrow \mathbb{R}_{>0}$ be a positive weight function on links. The boundary measure of an operational region R is*

$$v(\partial R) = \sum_{\ell \in \partial R} v(\ell). \quad (2.5)$$

We require v to be additive over disjoint link sets and monotone under inclusion.

Definition 4 (Holographic Density). *A constant $B > 0$ is called the holographic density of the QHIS. It specifies the maximal amount of accessible information (in bits) per unit of boundary measure. The capacity of a region R is*

$$I(R) = B v(\partial R). \quad (2.6)$$

The holographic capacity bound is the central structural postulate of the QHIS framework. It limits the informational complexity of any operational region purely in terms of its incidence boundary.

Axiom 1 (Holographic Capacity Bound). *For any operational region $R \subseteq P$ and any global state ρ_{global} on $\mathcal{H}_{\text{global}}$, the reduced state*

$$\rho_R = \text{Tr}_{P \setminus R}(\rho_{\text{global}}) \quad (2.7)$$

satisfies

$$S(\rho_R) \leq I(R), \quad (2.8)$$

where S denotes the von Neumann entropy.

Remark 1. *The holographic capacity bound is taken as a structural postulate of the QHIS framework. Under coarse-graining and classicalization, the boundary measure $v(\partial R)$ maps to geometric area, and the bound reduces to a familiar holographic area law. In this sense, the QHIS formulation provides a pre-geometric generalization of holographic entropy bounds and offers a discrete analogue of the entanglement-based mechanisms underlying emergent geometry in holographic dualities.*

These definitions provide the mathematical foundation for the observer construction in Sec. 3, the capacity bounds of Sec. 4, and the periodic cosmologies developed in Sec. 5.

3 Observers as Informational Patterns in QHIS

In conventional cosmology, the notion of an “observer” is typically treated as an external idealization: a coarse-grained subsystem with sufficient internal structure to register, store, and process information. Within the QHIS framework, however, observers must be described *internally* as patterns of informational degrees of freedom distributed across a finite set of hodons. This perspective is closely aligned with operational and information-theoretic approaches to quantum gravity, in which observers are not added to the theory but emerge as dynamically coherent substructures of the underlying substrate. The present section develops an operational notion of an observer in QHIS, identifies a minimal informational threshold for observerhood, and contrasts it with the vastly larger complexity required for a Boltzmann Brain-level observer. These distinctions play a central role in the capacity-limited cosmology constructed in later sections.

3.1 Observers as structured QHIS subregions

Let $R \subset P$ be an operationally defined subset of hodons. The quantum state accessible to an observer confined to R is the reduced density operator (2.7). In QHIS, an observer is not identified with a single hodon, but with a *structured pattern* of correlations distributed across many hodons. This reflects the broader view, common in quantum-gravitational and quantum-informational approaches, that observers correspond to dynamically stable, causally-embedded subsystems rather than fundamental entities.

A QHIS observer is characterized by:

1. **Internal differentiation:** the ability to realize at least two distinguishable internal states corresponding to different observational outcomes.
2. **Memory:** stable correlations among hodons in R that encode a short history of past inputs.
3. **Update dynamics:** conditional dependence of future internal states on both internal memory and incoming signals through the incidence boundary ∂R .
4. **Causal coherence:** compatibility of the above structures with the directed incidence relations and causal weights of the QHIS, ensuring that the observer is embedded consistently within the underlying causal substrate.

These criteria ensure that an observer in QHIS is a physically realizable, causally coherent subsystem capable of registering, processing, and responding to information. In this sense, QHIS provides a discrete analogue of the “observer as subsystem” paradigm familiar from tensor-network models, quantum error-correcting codes, and entanglement-based approaches to emergent geometry.

3.2 Minimal observer complexity

The informational complexity of an observer is quantified by the von Neumann entropy of its internal degrees of freedom:

$$S_{\text{obs}} \sim S(\rho_R). \quad (3.1)$$

A *minimal observer* is a region R_{min} whose internal structure supports:

- discrimination among a small number of input patterns,
- a short-term memory of past inputs,
- and conditional update rules.

Such functionality requires a nonzero informational footprint,

$$S_{\text{obs,min}} \sim S(\rho_{R_{\text{min}}}). \quad (3.2)$$

The precise value of $S_{\text{obs,min}}$ depends on the architecture of R_{min} , but toy models (see Sec. 6) indicate that only a few bits of sensory differentiation, a few bits of memory, and a minimal control structure are required. Thus,

$$S_{\text{obs,min}} \sim 5\text{--}10 \text{ bits}, \quad (3.3)$$

a finite and typically small scale.

In a quantum-gravitational context, this threshold plays a role analogous to the minimal entanglement or correlation structure required for a subsystem to behave as an “agent” or “probe” of the underlying geometry. QHIS provides a discrete, operational instantiation of this idea.

3.3 Boltzmann Brain-level observers

A Boltzmann Brain (BB) is a highly complex, low-entropy fluctuation containing:

- a large number of internal degrees of freedom,
- extensive memory and self-referential structure,
- and a sufficiently rich internal environment to support coherent self-awareness.

In informational terms, a BB requires an entropy scale S_{BB} that is astronomically large. Conservative physical estimates place its informational cost at

$$S_{\text{BB}} \sim 10^{20} - 10^{40} \text{ bits}, \quad (3.4)$$

or more [21, 2, 1]. Thus,

$$S_{\text{BB}} \gg S_{\text{obs,min}}, \quad (3.5)$$

a scale separation of many orders of magnitude.

From the perspective of QHIS, this separation is crucial: the holographic capacity bound of Sec. 4 implies that only regions with sufficiently large incidence boundaries could, in principle, support BB-level observers. In a capacity-limited QHIS cosmology, no such regions exist. This provides a structural, rather than dynamical, exclusion of BBs and aligns with the broader theme in quantum gravity that the entanglement structure of a region constrains the types of subsystems it can support.

4 Capacity-Limited Regions and Structural Exclusion of Boltzmann Brains

The QHIS framework imposes a finite informational capacity on every operationally defined region of the universe. This section develops the consequences of this constraint for the existence of observers. Under mild assumptions, the holographic capacity bound implies a strict upper limit on the complexity of any observer that can be supported within a region R . When this upper limit lies only slightly above the minimal threshold for observerhood, Boltzmann Brain-level observers become *structurally impossible*. This yields a capacity-based resolution of the BB problem that does not rely on dynamical timescale arguments and is naturally aligned with the broader theme in quantum gravity that entanglement structure constrains the types of subsystems a region can support.

4.1 Holographic capacity of an operational region

Let $R \subset P$ be an operationally defined subset of hodons. The incidence boundary ∂R is the minimal set of links whose removal eliminates all positive-propensity paths between R and its complement. In QHIS, the information accessible to observers confined to R is determined not by its volume or temporal duration, but by the combinatorial structure of its boundary, as encoded in the capacity definition (Eq. (2.6)).

Crucially, the capacity $I(R)$ is *finite* for every operational region. No subsystem of the universe can support arbitrarily large amounts of structured information, regardless of cosmological timescales. This finiteness is a structural feature of the QHIS substrate and does not depend on any semiclassical approximation.

The capacity $I(R)$ may be viewed as a discrete, pre-geometric analogue of the Bousso entropy bound. Whereas the semiclassical bound constrains the entropy passing through light-sheets of codimension-two surfaces, the QHIS bound constrains the entropy accessible to any operational region purely through its incidence boundary. Under coarse-graining, $v(\partial R)$ maps to geometric area, and $I(R)$ reproduces an area-law scaling consistent with holography. In this sense, the QHIS capacity bound provides a combinatorial precursor to the generalized entropy functional appearing in quantum extremal surface and island constructions [22].

4.2 Informational requirements for observerhood

Let $S(\rho_R)$ denote the von Neumann entropy of the reduced state accessible to an observer confined to R . Combining the minimal-observer requirement $S_{\text{obs,min}} \leq S_{\text{obs}}$ with the holographic capacity bound stated in Axiom 1 yields

$$S_{\text{obs,min}} \lesssim S(\rho_R) \leq I(R). \quad (4.1)$$

Equation (4.1) defines an “observer window” for each region R : the range of informational complexities that can be supported by the region’s boundary capacity. If $I(R)$ is only slightly larger than $S_{\text{obs,min}}$, then R can support minimal or near-minimal observers but cannot support observers of significantly greater complexity.

From a quantum gravity perspective, this window plays a role analogous to the entanglement constraints that determine which subsystems can emerge as semiclassical observers in holographic or tensor-network models. The QHIS formulation makes this constraint explicit in a pre-geometric setting.

4.3 Boltzmann Brains and capacity thresholds

In semiclassical cosmology, the enormous Hilbert-space dimension of a de Sitter horizon patch permits a BB in principle, albeit with extremely small probability. In QHIS, however, the holographic capacity bound implies that a BB can exist in R only if

$$I(R) \geq S_{\text{BB}}. \quad (4.2)$$

If no operational region satisfies Eq. (4.2), then BB-level observers are *structurally excluded*: the informational substrate of the universe simply does not contain regions with sufficient capacity to support the correlations required for a BB. This exclusion is independent of global dynamics. Even if the universe were to persist for arbitrarily long times, or even if the global state were ergodic, BBs could not arise in any region whose capacity lies below S_{BB} .

This structural exclusion mirrors the logic of holographic entropy bounds in semiclassical gravity: certain high-complexity configurations are forbidden not because they are dynamically improbable, but because they violate fundamental capacity constraints of the underlying substrate.

4.4 Capacity-limited cosmology

As established in Sec. 3, the scale separation $S_{\text{BB}} \gg S_{\text{obs,min}}$ is enormous. A cosmology is therefore *capacity-limited* if every operational region R satisfies

$$S_{\text{obs,min}} \lesssim I(R) \ll S_{\text{BB}}. \quad (4.3)$$

In such a universe:

- ordinary observers can exist, since their informational complexity fits within the available capacity;
- BB-level observers cannot exist, since no region has sufficient capacity to support them;
- the BB problem is resolved *structurally*, not dynamically.

When combined with a commensurable-spectrum Hamiltonian (Sec. 5), capacity-limited cosmology yields a universe that is periodic, non-ergodic, and free of BBs in principle. The global state cycles through a finite sequence of macrostates, each respecting the capacity constraints of Eq. (4.3). The BB problem is thereby reframed as a question about the *granularity of observers* permitted by the informational substrate of the universe.

From the perspective of quantum gravity, this suggests that the entanglement and correlation structure of the underlying substrate—not geometric volume or temporal duration—determines which observers are physically realizable. QHIS thus provides a discrete, operational setting in which structural entropy bounds directly constrain the space of admissible observers.

5 Periodic QHIS Cosmology

This section develops a cosmological model in which the global QHIS state evolves periodically under a commensurable-spectrum Hamiltonian. The resulting universe is non-ergodic, cyclic, and capacity-limited: it traverses a finite sequence of macrostates, each of which respects the holographic capacity constraints of Sec. 4. Such a model naturally supports ordinary observers while structurally excluding Boltzmann Brains. We also construct a capacity-limited Page curve that arises without reference to semiclassical geometry or quantum extremal surfaces, illustrating how finite correlation capacity alone can reproduce the qualitative features of black-hole evaporation.

From a quantum gravity perspective, the periodic evolution generated by a commensurable-spectrum Hamiltonian provides a discrete analogue of relational or emergent time: the “clock” of the universe is encoded in its internal correlations rather than in an external time parameter. This aligns with approaches in which global time is not fundamental but arises from the dynamics of subsystems.

5.1 Commensurable-spectrum Hamiltonian on a QHIS

Let us consider $\mathcal{H}_{\text{global}}$, the global Hilbert space as specified in Eq. (2.3).

Choose a finite set of orthonormal macrostates $\{|n\rangle\}_{n=0}^{M-1} \subset \mathcal{H}_{\text{global}}$, representing coarse-grained QHIS states at successive phases of a cosmological cycle (e.g. bounce, expansion, structure formation, de Sitter-like epoch, purification).

Define Fourier modes

$$|\phi_k\rangle = \frac{1}{\sqrt{M}} \sum_{n=0}^{M-1} e^{2\pi i k n / M} |n\rangle, \quad k = 0, \dots, M-1. \quad (5.1)$$

A commensurable-spectrum Hamiltonian is then given by

$$H_{\text{QHIS}} = \sum_{k=0}^{M-1} E_k |\phi_k\rangle\langle\phi_k| + H_{\perp}, \quad (5.2)$$

where $H_{\perp}$ acts on the orthogonal complement of $\text{span}\{|n\rangle\}$ and

$$E_k = \frac{2\pi k}{MT_{\star}}. \quad (5.3)$$

Lemma 1. *If $E_k = 2\pi k/(MT_\star)$ for $k = 0, \dots, M-1$, then*

$$e^{-iE_k T_\star} = 1 \quad \text{for all } k,$$

and therefore the global evolution is exactly periodic with period $T_\star$:

$$U(T_\star) = e^{-iH_{\text{QHIS}} T_\star} = \mathbb{I} \quad \text{on } \text{span}\{|n\rangle\}.$$

Proof. For each Fourier mode,

$$e^{-iE_k T_\star} = \exp\left[-i \frac{2\pi k}{MT_\star} T_\star\right] = e^{-2\pi i k/M}.$$

Since k is an integer and $e^{-2\pi i k/M}$ is an M th root of unity, we have $e^{-2\pi i k/M} = 1$ for all k . Thus every eigenvalue of $e^{-iH_{\text{QHIS}} T_\star}$ on $\text{span}\{|n\rangle\}$ is unity, and the evolution operator reduces to the identity on this subspace. The action of $H_\perp$ does not affect periodicity, as it acts on the orthogonal complement. $\square$

Remark 2. *The construction above is the discrete Fourier transform of a cyclic shift operator. The states $|n\rangle$ form a basis for an M -cycle, and the Fourier modes $|\phi_k\rangle$ diagonalize the shift. The commensurable spectrum ensures that the evolution operator returns exactly to the identity after time $T_\star$, yielding an exactly periodic cosmology with M distinct macrostates. This structure is analogous to cyclic tensor-network models and to discrete-time approaches in which the global state evolves along a closed trajectory in Hilbert space.*

5.2 Non-ergodic evolution on a one-dimensional orbit

Choose the initial state to be one of the macrostates, $|\Psi(0)\rangle = |0\rangle$. At discrete times $t_n = n\Delta t$ with $\Delta t = T_\star/M$, the state evolves as

$$|\Psi(t_n)\rangle = |n\rangle, \quad n = 0, \dots, M-1, \quad (5.4)$$

with $|M\rangle \equiv |0\rangle$. The global state thus traces a one-dimensional, non-ergodic orbit in Hilbert space, revisiting the same sequence of macrostates every cycle. No part of the Hilbert space outside this orbit is ever explored, and the universe never enters high-entropy configurations beyond those encoded in the chosen macro-basis.

This non-ergodicity is essential: it ensures that the universe does not wander into regions of Hilbert space that would require informational capacities exceeding the holographic bounds of Sec. 4. In quantum gravity terms, the evolution is confined to a “capacity-safe submanifold” of Hilbert space, analogous to how semiclassical spacetimes occupy a restricted subset of states in holographic theories.

5.3 Capacity-limited Page curves

Each macrostate $|n\rangle$ corresponds to a coarse-grained QHIS configuration with a well-defined partition into interior, radiation, and environment regions. Let R_{rad} denote the radiation region. The radiation entropy at phase n is

$$S_{\text{rad}}(n) = S(\text{Tr}_{P \setminus R_{\text{rad}}}(|n\rangle\langle n|)). \quad (5.5)$$

Because each $|n\rangle$ is constructed to respect the holographic capacity bound,

$$S_{\text{rad}}(n) \leq I(R_{\text{rad}}) = B v(\partial R_{\text{rad}}), \quad (5.6)$$

the radiation entropy cannot exceed the boundary capacity of the region.

By choosing the macrostates appropriately, one obtains a capacity-limited Page curve:

- early-time growth of $S_{\text{rad}}(n)$ as new correlations cross the incidence boundary,
- saturation at a maximum value determined by $Bv(\partial R_{\text{rad}})$,
- and late-time purification as the interior region loses independent degrees of freedom.

This Page-curve-like behavior arises without reference to semiclassical geometry, quantum extremal surfaces, or island formulas. It follows directly from the finite correlation capacity of incidence boundaries in QHIS. In this sense, the QHIS capacity bound acts as a discrete, pre-geometric analogue of the generalized entropy functional in the QES/island framework: saturation reflects the maximal correlation capacity of the boundary, and purification reflects the loss of independent interior degrees of freedom.

5.4 Periodic, capacity-limited cosmology

The combination of commensurable-spectrum dynamics and holographic capacity bounds yields a cosmology with the following properties:

1. **Exact periodicity:** the universe revisits the same sequence of macrostates every period T_* .
2. **Non-ergodicity:** the global state is confined to a one-dimensional orbit in Hilbert space.
3. **Capacity limitation:** every macrostate $|n\rangle$ respects the holographic bound

$$S_R(n) \leq I(R) = Bv(\partial R) \quad (5.7)$$

for all operational regions R , where $S_R(n)$ denotes the entropy of the reduced state on R at phase n .

4. **Structural BB exclusion:** no region has sufficient capacity to support a Boltzmann Brain-level observer.

In such a universe, ordinary observers arise naturally as structured patterns within regions whose capacities lie just above the minimal observer threshold. Boltzmann Brains, by contrast, are excluded in principle: the informational substrate of the universe does not contain regions with sufficient capacity to support them, and the global dynamics never explores states beyond the capacity-limited orbit.

The periodic evolution generated by a commensurable-spectrum Hamiltonian is consistent with relational or discrete-time approaches to quantum gravity, where global time is not fundamental but emerges from internal correlations. In this sense, the periodic QHIS cosmology may be interpreted as a discrete analogue of a closed timelike trajectory in Hilbert space, constrained by informational rather than geometric considerations. The model illustrates how a pre-geometric substrate with finite correlation capacity can give rise to cyclic cosmological behavior without invoking semiclassical geometry.

In the next section, we construct explicit toy models that illustrate these ideas in detail.

6 Explicit Toy Models

To illustrate the mechanisms developed in the preceding sections, we now construct three explicit toy models: (i) a minimal observer supported by a small hodon region, (ii) a capacity-limited Page curve arising from finite incidence-boundary capacity, and (iii) a small- M periodic QHIS cosmology implementing a commensurable-spectrum Hamiltonian. These examples demonstrate concretely how ordinary observers can exist while Boltzmann Brains are structurally excluded.

6.1 A 16-hodon minimal observer

Consider an operational region

$$R = \{p_1, \dots, p_{16}\}, \quad (6.1)$$

with each hodon carrying a qubit, $\mathcal{H}_{p_i} \cong \mathbb{C}^2$. The Hilbert space of the region is therefore

$$\mathcal{H}_R \cong (\mathbb{C}^2)^{\otimes 16}, \quad \dim \mathcal{H}_R = 2^{16}. \quad (6.2)$$

To model a minimal observer, we partition R into functional layers:

- **Sensory layer:** $p_1, \dots, p_4$ (4 bits of input).
- **Memory layer:** $p_5, \dots, p_{12}$ (8 bits of memory).
- **Control layer:** $p_{13}, \dots, p_{16}$ (4 bits for conditional update rules).

A minimal observer requires only a few bits of sensory differentiation, a few bits of memory, and a small control structure. A reasonable estimate for the minimal informational complexity is therefore

$$S_{\text{obs,min}} \approx 10 \text{ bits}. \quad (6.3)$$

Let the incidence boundary ∂R consist of five links, each contributing one unit to the boundary measure:

$$v(\partial R) = 5. \quad (6.4)$$

With holographic density $B = 2$ bits per unit, the capacity bound is

$$I(R) = B v(\partial R) = 10 \text{ bits}. \quad (6.5)$$

Thus R can support a minimal observer but cannot support significantly more complex structures. In particular, since $S_{\text{BB}} \gg 10$ bits, no BB-level observer can exist in R (see Fig. 1).

From a quantum gravity viewpoint, this example provides a discrete analogue of the entanglement-based constraints that determine which subsystems can emerge as “observers” in tensor-network or holographic models. The incidence boundary plays the role of a minimal cut, and the capacity bound acts as a combinatorial precursor to an area-law constraint.

6.2 A capacity-limited Page curve

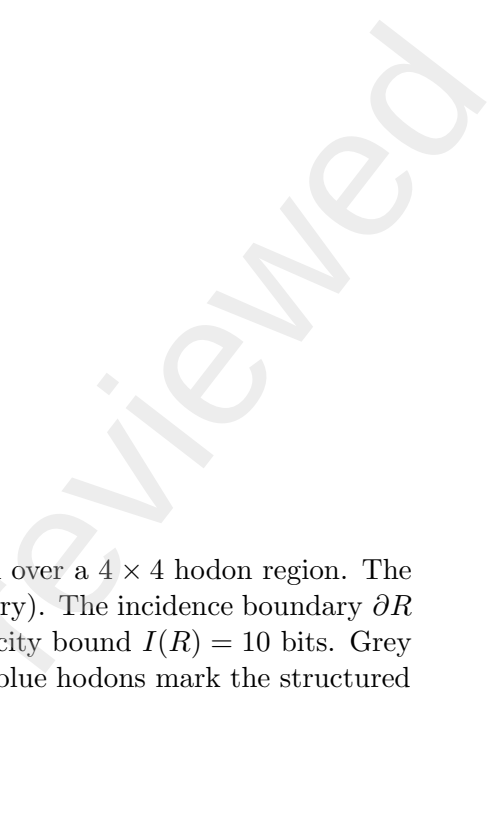
Let R_{rad} be a radiation region with boundary capacity

$$I(R_{\text{rad}}) = B v(\partial R_{\text{rad}}) = 10 \text{ bits}. \quad (6.6)$$

We construct a Page-curve-like entropy profile along a sequence of macrostates $|n\rangle$, $n = 0, \dots, 5$, representing successive phases of a cosmological cycle. Define $S_{\text{rad}}(n)$ following Eq. (5.5). Choose the values

$$S_{\text{rad}}(n) = \{0, 4, 8, 9, 6, 2\} \text{ bits}. \quad (6.7)$$

This profile exhibits:



- early-time growth ($0 \rightarrow 4 \rightarrow 8$),
- saturation near the capacity limit ($8 \rightarrow 9$),
- and late-time purification ($9 \rightarrow 6 \rightarrow 2$).

- The holographic bound is respected at every step:

This demonstrates that Page-curve-like behavior can arise purely from capacity-limited information flow across incidence boundaries, without reference to semiclassical geometry or quantum extremal surfaces (see Fig. 2).

6.3 A small- M periodic QHIS cosmology

$$|\phi_k\rangle = \frac{1}{\sqrt{6}} \sum_{n=0}^5 e^{2\pi i k n/6} |n\rangle, \quad k = 0, \dots, 5. \quad (6.9)$$
$$H_{\text{QHIS}} = \sum_{k=0}^5 \frac{2\pi k}{6T_{\star}} |\phi_k\rangle\langle\phi_k| + H_{\perp}, \quad (6.10)$$

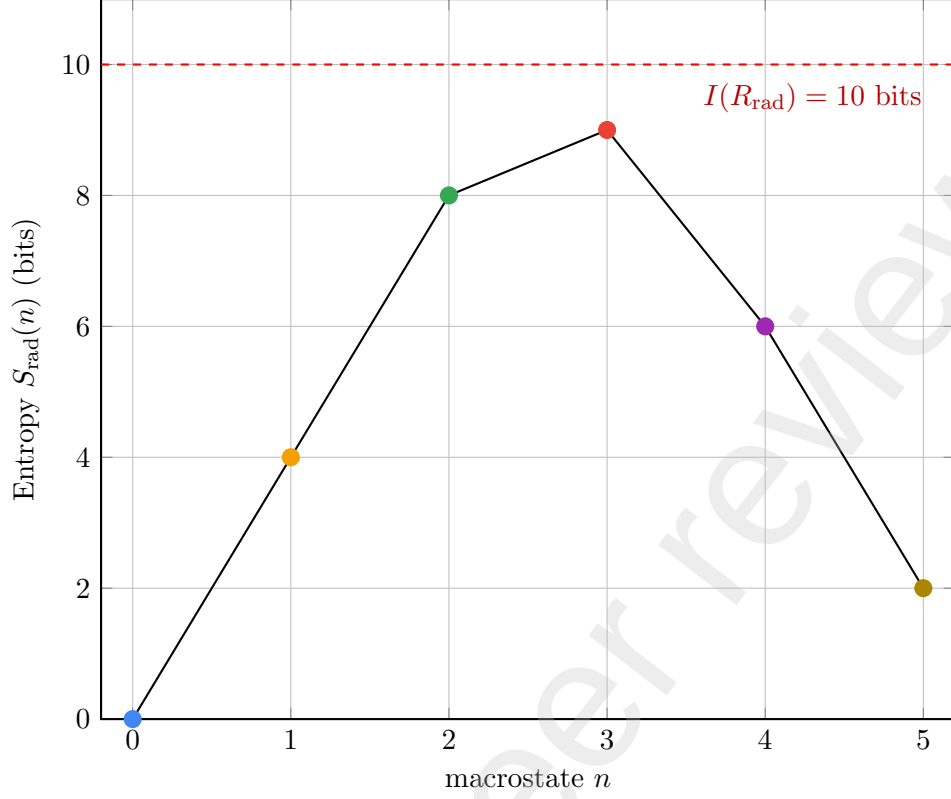


Figure 2: Capacity-limited Page curve for the toy model with boundary capacity $I(R_{\text{rad}}) = 10$ bits. Each point is color-coded to match the corresponding macrostate in Fig. 3. The entropy profile $S_{\text{rad}}(n) = \{0, 4, 8, 9, 6, 2\}$ shows early-time growth, near-saturation, and late-time purification.

so that

$$U(T_\star) = e^{-iH_{\text{QHIS}}T_\star} = \not\llcorner \quad \text{on } \text{span}\{|n\rangle\}_{n=0}^5. \quad (6.11)$$

With initial state $|\Psi(0)\rangle = |0\rangle$, the evolution is

$$|\Psi(n\Delta t)\rangle = |n\rangle, \quad \Delta t = T_\star/6. \quad (6.12)$$

The universe cycles through the six macrostates in a strictly periodic, non-ergodic manner.

By choosing each $|n\rangle$ to satisfy the capacity constraints of Sec. 4, we obtain a cosmology in which:

- ordinary observers fit within the allowed informational budgets of their regions,
- the radiation entropy follows the capacity-limited Page curve constructed above,
- and BB-level observers are structurally excluded, since no region satisfies $I(R) \geq S_{\text{BB}}$.

This explicit construction demonstrates how periodicity, non-ergodicity, and holographic capacity bounds combine to yield a universe in which ordinary observers arise naturally while Boltzmann Brains are impossible in principle (see Fig. 3).

From a quantum gravity perspective, the model provides a discrete analogue of a closed time-like trajectory in Hilbert space, with the “arrow of time” emerging from the cyclic ordering of macrostates rather than from geometric expansion or contraction. The capacity constraints ensure that the evolution remains confined to a “holographically safe” subset of Hilbert space, analogous to how semiclassical spacetimes occupy a restricted region of the holographic state space.

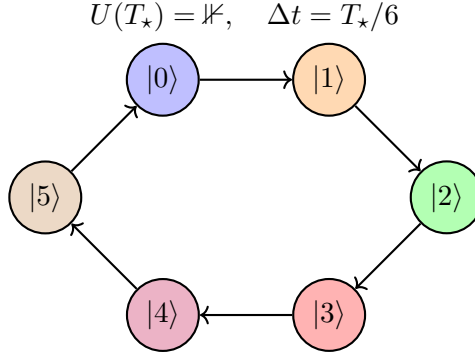


Figure 3: A periodic QHIS cosmology with $M = 6$ macrostates. Each macrostate $|n\rangle$ is color-coded to match the corresponding macrostates in Fig. 2. The universe evolves deterministically through the cycle $|0\rangle \rightarrow |1\rangle \rightarrow \cdots \rightarrow |5\rangle \rightarrow |0\rangle$, repeating every period $T_\star$. Each state satisfies the capacity constraints of Sec. 4, ensuring that ordinary observers fit within the allowed informational budgets while BB-level observers remain structurally excluded.

7 Discussion

The constructions developed in this work suggest a shift in how cosmology may be conceptualized when informational structure, rather than geometric structure, is taken as fundamental. In a QHIS-based universe, the capacity of an operational region is determined not by its spatial volume or semiclassical horizon area, but by the combinatorial properties of its incidence boundary. This shift has significant implications for the Boltzmann Brain problem, the emergence of observers, and the interpretation of cosmological dynamics. It also aligns naturally with quantum gravity approaches in which causal structure, entanglement, and information flow are more primitive than geometry.

The capacity-limited Page-curve-like behavior observed in QHIS is closely related to entanglement-entropy dynamics in semiclassical gravity. In holographic settings, the Page curve arises from the competition between quantum extremal surfaces and island contributions. In QHIS, an analogous saturation and purification emerge from the finite correlation capacity across incidence boundaries, without invoking extremal surfaces or bulk geometry. This suggests that the QHIS capacity bound may serve as a discrete precursor to the generalized entropy functional appearing in the QES/island formula, with $v(\partial R)$ playing the role of a pre-geometric “area” and $S(\rho_R)$ representing the accessible entropy of the region.

7.1 Structural versus dynamical avoidance of Boltzmann Brains

Most existing approaches to the Boltzmann Brain problem rely on *dynamical* arguments: the rate of BB nucleation, the duration of de Sitter phases, or the global measure on cosmological histories. Carroll’s commensurable-spectrum proposal refines this strategy by restricting the universe to a periodic, non-ergodic orbit in Hilbert space. In such a model, BBs remain possible in principle, but the universe never evolves long enough in any direction for them to appear with appreciable probability.

The QHIS framework provides a qualitatively different mechanism. The holographic capacity bound $S(\rho_R) \leq I(R)$ limits the informational complexity of any region R . If this capacity lies only slightly above the minimal threshold for observerhood $S_{\text{obs,min}}$, then BB-level observers are *structurally impossible*: their informational requirements exceed the maximal capacity of any region, regardless of cosmological timescales or global dynamics. This yields a *structural* resolu-

tion of the BB problem, independent of dynamical considerations. In this sense, QHIS provides a pre-geometric analogue of the idea that certain high-complexity configurations are forbidden by holographic entropy bounds.

7.2 Observer granularity as a physical parameter

A central conceptual contribution of this work is the recognition that the *granularity* of observers is a physical parameter of the universe. In a capacity-limited QHIS cosmology, the informational substrate supports only observers whose internal complexity fits within the boundary capacity of the regions they inhabit. Minimal or near-minimal observers are permitted, whereas highly complex, self-contained observers such as Boltzmann Brains are not.

Demonstrating that ordinary observers naturally emerge within regions whose capacities lie only slightly above the minimal observer threshold is therefore essential. It shows that the universe is not sterile: it supports structured, causally embedded, memory-bearing observers like ourselves. At the same time, the same capacity constraints prevent the formation of BB-level observers, whose internal correlation structure would require informational resources far exceeding any region's capacity. Thus, the universe supports “small” observers but forbids “large” ones, and this asymmetry is built into the informational substrate itself.

7.3 Relation to holography and emergent geometry

The capacity-limited nature of QHIS regions echoes familiar themes in holography, where the entropy of a region is bounded by the area of its boundary. However, the QHIS bound is more primitive: it applies even in non-geometric or pre-geometric regimes, and it is defined purely in terms of incidence relations and causal weights. Geometry emerges only after coarse-graining, and the holographic bound survives this transition.

This suggests that structural exclusion of BBs may be a generic feature of holography-inspired theories of quantum gravity. If the fundamental substrate enforces finite informational capacity for all operational regions, then BB-level observers may be excluded in a wide class of models, not only in QHIS. The framework also resonates with entanglement-based approaches to emergent geometry, where the structure of correlations determines effective spatial relations.

7.4 Non-ergodicity, Page-curve-like behavior, and the role of periodicity

The commensurable-spectrum Hamiltonian plays a complementary role. While capacity constraints alone can exclude BBs, periodicity ensures that the global state remains confined to a finite set of macrostates, each of which respects the capacity bounds. This prevents the universe from exploring exotic or pathological configurations that might arise in a fully ergodic evolution.

Equally important is the emergence of Page-curve-like behavior from finite correlation capacity across incidence boundaries. In semiclassical gravity, the Page curve reflects the fact that a region cannot sustain arbitrarily large entanglement: correlations grow, saturate at a boundary-determined maximum, and then purify. In QHIS, the same qualitative behavior arises without reference to geometry or quantum extremal surfaces. Instead, it emerges from the finite correlation capacity imposed by the holographic bound.

This correspondence is structural rather than analogical. The Page-curve-like profile is the *dynamical fingerprint* of the same structural constraint that forbids BBs. If correlations across ∂R cannot grow beyond a small, boundary-determined maximum, then no region can accumulate the vast internal correlation structure required for a BB. Thus, the capacity-limited Page curve and

the structural exclusion of BBs are two manifestations of a single underlying principle: the universe has finite informational granularity.

7.5 Implications for cosmological naturalness

Capacity-limited periodic cosmology offers a new angle on the problem of cosmological naturalness [23, 24]. In standard cosmology, the low-entropy initial state of the universe is often regarded as highly fine-tuned. In the present framework, however, the global state is not selected from a measure over initial conditions but is determined by the periodic orbit of the commensurable-spectrum Hamiltonian. The “initial” state is merely one point on a closed trajectory. Moreover, the informational capacity of regions constrains the complexity of macrostates, preventing the universe from entering high-entropy configurations that would otherwise dominate.

7.6 Limitations and future directions

The toy models presented here are highly idealized. A full QHIS cosmology would require:

- a detailed specification of the incidence structure and causal weights,
- a derivation of the holographic density B from more fundamental principles,
- and a demonstration that realistic cosmological dynamics can emerge from QHIS under coarse-graining.

Nevertheless, the examples illustrate that capacity-limited periodic cosmology is a viable framework for addressing the BB problem and may shed light on deeper questions about the informational foundations of spacetime. It would be valuable to explore whether QHIS admits a continuum limit compatible with semiclassical gravity, whether its capacity bound can be related directly to generalized entropy in the QES/island framework, and whether its periodic dynamics can be embedded in a broader class of relational-time or discrete-time quantum gravity models.

In the next section we summarize the main conclusions and outline directions for further research.

8 Conclusion

The Boltzmann Brain problem has long been regarded as a deep challenge for cosmological models containing long-lived de Sitter phases or large Hilbert spaces. In such settings, exceedingly rare entropy-lowering fluctuations eventually dominate over ordinary observers, threatening the empirical coherence of cosmology by rendering our observational situation atypical. Carroll’s proposal of a finite-dimensional, commensurable-spectrum universe offers a dynamical loophole: exact periodicity and non-ergodicity restrict the effective duration of high-entropy epochs, suppressing BB formation by limiting the universe’s exploration of Hilbert space.

In this work we have shown that, within the QHIS framework, a sharper and more structural resolution is available. QHIS models impose a holographic counting bound on every operational region, ensuring that its accessible informational capacity is finite and determined by the combinatorial properties of its incidence boundary. When combined with a commensurable-spectrum global Hamiltonian, this constraint yields a cosmology that is periodic, non-ergodic, and fundamentally *capacity-limited*. Ordinary observers, whose informational complexity is modest, fit naturally within the allowed capacities of suitable regions. In contrast, Boltzmann Brain-level observers require correlation structures far exceeding the capacity of any region and are therefore *impossible*.

in principle. The BB problem is thus reframed as a question about the informational granularity of the universe rather than its temporal duration or dynamical history.

A key insight of this work is that Page-curve-like behavior emerges naturally in QHIS from finite correlation capacity across incidence boundaries. In semiclassical gravity, the Page curve reflects the inability of a region to sustain arbitrarily large entanglement. In QHIS, the same qualitative behavior arises without reference to geometry, horizons, or quantum extremal surfaces. The Page-curve-like profile becomes the operational signature of finite informational capacity, and the same mechanism that limits entanglement growth also limits the complexity of observers. This unifies two phenomena—entropy saturation and observer granularity—under a single structural principle.

We illustrated these ideas with explicit toy models: a minimal observer supported by a small hodon region, a capacity-limited Page curve, and a small- M periodic QHIS cosmology. These examples demonstrate that ordinary observers arise naturally within the informational budgets of capacity-limited regions, while BB-level observers cannot be realized in any region of the universe. They also show how Page-curve-like behavior can emerge from purely informational constraints, independent of semiclassical geometry.

The broader implication is that the informational structure of the universe may be more fundamental than its geometric structure. If the substrate of the universe enforces finite informational capacity for all operational regions, then the space of physically realizable observers is sharply constrained. This perspective suggests new avenues for addressing cosmological naturalness, the emergence of low-entropy initial conditions, and the foundations of holography.

Several open questions remain. A full QHIS cosmology requires a detailed specification of the incidence structure and causal weights, a deeper understanding of the holographic density B , and a demonstration that realistic cosmological dynamics can emerge under coarse-graining. It would also be valuable to explore whether capacity-limited periodic cosmology can be embedded within or derived from more complete theories of quantum gravity.

Nevertheless, the results presented here indicate that capacity-limited periodic cosmology offers a promising and conceptually robust framework in which ordinary observers arise naturally and Boltzmann Brains are excluded by the fundamental informational structure of the world. This structural resolution highlights the central role of informational capacity in shaping the space of possible observers and points toward a deeper informational foundation for cosmology and quantum gravity.

These results suggest that informational capacity may play a structural role in quantum gravity, analogous to geometric area in semiclassical holography. The QHIS framework therefore offers a potential bridge between discrete, pre-geometric models and entanglement-based descriptions of emergent spacetime. Further work is needed to investigate whether QHIS admits a continuum limit compatible with semiclassical gravity and whether its capacity bound can be related directly to generalized entropy in the QES/island framework.

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